

KIARASH ALIREZAEI

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EDUCATION

University of Toronto, St. George

Toronto, Ontario

Bachelor of Applied Science in Computer Engineering (GPA: 3.76/4.0)

Sep 2022 – May 2027

Minors and Certificates: Artificial Intelligence Engineering Minor, Engineering Business Minor, Cybersecurity Certificate

Relevant Coursework: Algorithms and Data Structures, Operating Systems (C), Distributed Systems (Go), Computer Networks, Introduction to Machine Learning, Introduction to Artificial Intelligence, Applied Fundamentals of Deep Learning (Python), Probability & Applications, Computer Security, Digital Systems (Verilog), Digital Electronics, Computer Organization (Assembly)

EXPERIENCE

IBM Canada

May 2025 – Aug 2026

Compiler Developer Intern – IBM Python for z/OS

Markham, Ontario

- Developed and enhanced IBM Open Enterprise SDK for Python using C and Python, debugging compiler and runtime issues, integrating newly introduced z/OS system functions into CPython, and improving compatibility across Python builds.
- Investigated customer-reported issues involving Python imports, third-party package installation, application debugging, and underlying compiler or runtime defects, while contributing fixes to upstream Python projects such as distutils.
- Developed automation scripts to download, install, and validate pip packages from the IBM Python AI Toolkit on z/OS, improving package compatibility testing and validation workflows.
- Integrated upstream libraries such as zstd and maintained automated test and CI pipelines using Git and Buildbot to support reliable Python builds and regression testing on z/OS.
- Authored customer-focused technical blogs covering Python libraries, development tools, package installation, debugging, and common IBM Open Enterprise SDK for Python workflows.

Reviewerly

May 2024 – Aug 2024

Software Engineer Intern

Toronto, Ontario

- Developed RESTful APIs and backend functionality using Java and MongoDB, supporting data creation, retrieval, and application workflows.
- Migrated search functionality from MongoDB queries to Apache Solr, improving search performance and response times by over 100%.
- Built responsive React interfaces and integrated backend APIs using React Query, improving state management, data visualization, and user interaction.

PROJECTS

Fault-Tolerant Distributed Key-Value Store | Go, Raft, RPC, Distributed Systems

Jan 2025 - Apr 2025

- Implemented the Raft consensus protocol in Go, including randomized leader election, heartbeat-based failure detection, replicated logs, majority-based commits, and automatic recovery from node failures.
- Developed persistent state and optimized log-reconciliation mechanisms to recover crashed replicas, resolve conflicting logs, and efficiently resynchronize nodes after network partitions.
- Built a linearizable distributed key-value store on top of Raft, supporting concurrent Get/Put operations, client retries, leader changes, server restarts, and unreliable network conditions.

Skin Lesion Classifier (NLC7) | Python, PyTorch, Matplotlib, CNNs, Transfer Learning

May 2024 – Aug 2024

- Processed the large-scale HAM10000 dataset, applying advanced data augmentation techniques using Pandas and Torch libraries to balance 6000 images across cancer types and ensure robust model training.
- Designed a complex deep learning model incorporating U-Net for precise image segmentation, followed by ResNet-50 for feature extraction and MLP for classification, enabling accurate identification of skin cancer types.
- Researched state-of-the-art AI image processing techniques and applied hyperparameter tuning across U-Net, CNNs, and MLP models, achieving 92% accuracy while reducing RAM usage by 75% and optimizing model performance.

Interactive Maps Application Development | C++ (STL), EZGL, GTK

Feb 2024 – Mar 2024

- Implemented API calls for real-time data processing and visualization using C++ and the STL library, enabling dynamic user interactions and up-to-date map information.
- Collaborated on UI/UX design to create an intuitive, responsive interface for efficient navigation and interaction.
- Developed and integrated advanced pathfinding algorithms such as Multi-Dijkstra's and A*, along with a solution for the Traveling Salesman Problem, improving route optimization and overall performance.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Go, JavaScript, SQL, Bash/Shell, Verilog, Assembly, MATLAB

Systems & Platforms: CPython, z/OS USS, Linux, POSIX APIs, Compiler & Runtime Development, Distributed Systems, REST APIs

AI & Machine Learning: PyTorch, TensorFlow, NumPy, Pandas, Matplotlib, Deep Learning, Neural Networks, Transfer Learning, Computer Vision

Frameworks & Software: React, React Native, Spring Boot, Django, GTK, EZGL

Hardware & Digital Systems: FPGA (DE1-SoC), Quartus, ModelSim, VGA, PS/2, Finite State Machines, Datapaths, Digital Logic

Tools & DevOps: Git, Buildbot, Ansible, Docker, pytest, pip, virtual environments, Bandit, Detect-Secrets, IBM Clang, CPULator

Databases & Search: MongoDB, MySQL, Apache Solr

Core Competencies: Systems Programming, Concurrency, Fault Tolerance, RPC, CI/CD, Python Packaging, Test Automation, Debugging, API Development, Agile Development